FACE MASK DETECTION USING DEEP LEARNING

Mini Project Report

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Department: Computer Science & Engineering
Semester: 4th Semester
Academic Year: 2026

Certificate

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Acknowledgement

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Abstract

This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH MASK and WITHOUT MASK and provides visual and audible alerts. This project implements a real-time face mask detection system using a Convolutional Neural Network (CNN), OpenCV, and TensorFlow. It classifies faces into WITH

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Introduction

The COVID-19 pandemic highlighted the importance of automated monitoring systems. This project demonstrates a practical computer vision solution for detecting mask usage in real time. The COVID-19 pandemic highlighted the importance of automated monitoring systems. This project demonstrates a practical computer vision solution for detecting mask usage in real time. The COVID-19 pandemic highlighted the importance of automated monitoring systems. This project demonstrates a practical computer vision solution for detecting mask usage in real time. The COVID-19 pandemic highlighted the importance of automated monitoring systems. This project demonstrates a practical computer vision solution for detecting mask usage in real time. The COVID-19 pandemic highlighted the importance of automated monitoring systems. This project demonstrates a practical computer vision solution for detecting mask usage in real time. 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Literature Survey

Existing methods include traditional image processing and deep learning. CNN-based approaches generally provide better feature extraction and classification performance. Existing methods include traditional image processing and deep learning. CNN-based approaches generally provide better feature extraction and classification performance. Existing methods include traditional image processing and deep learning. CNN-based approaches generally provide better feature extraction and classification performance. Existing methods include traditional image processing and deep learning. CNN-based approaches generally provide better feature extraction and classification performance. Existing methods include traditional image processing and deep learning. CNN-based approaches generally provide better feature extraction and classification performance. Existing methods include traditional image processing and deep learning. 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Problem Statement

Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance. Manual monitoring of mask usage is inefficient. An automated vision-based system helps improve safety and compliance.

Objectives

Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference. Detect faces, classify mask usage, alert users without masks, and demonstrate real-time inference.

Existing System

Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent. Traditional manual inspection is time-consuming and inconsistent.

Proposed System

A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference. A CNN trained on mask images is integrated with OpenCV and a Haar Cascade face detector for live webcam inference.

Software and Hardware Requirements

Python 3.11, TensorFlow, OpenCV, NumPy, Matplotlib, Jupyter Notebook, Webcam, Windows PC.
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Python 3.11, TensorFlow, OpenCV, NumPy, Matplotlib, Jupyter Notebook, Webcam, Windows PC.

Dataset Description

The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized. The dataset contains two classes: with_mask and without_mask. Images are resized to 128x128 pixels and normalized.

Data Preprocessing

Images are loaded, resized, normalized, labels encoded, and split into training and testing sets.
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CNN Architecture

Three convolution blocks followed by pooling, flattening, dense layers, dropout, and softmax output.
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Implementation

The model is trained, saved, loaded, and used with webcam frames. Faces are detected and classified in real time. The model is trained, saved, loaded, and used with webcam frames. Faces are detected and classified in real time. The model is trained, saved, loaded, and used with webcam frames. Faces are detected and classified in real time. The model is trained, saved, loaded, and used with webcam frames. Faces are detected and classified in real time. The model is trained, saved, loaded, and used with webcam frames. Faces are detected and classified in real time. The model is trained, saved, loaded, and used with webcam frames. Faces are detected and classified in real time. The model is trained, saved, loaded, and used with webcam frames. Faces are detected and classified in real time. The model is trained, saved, loaded, and used with webcam frames. Faces are detected and classified in real time. The model is trained, saved, loaded, and used with webcam frames. 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Results

The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text. The project successfully displayed green boxes for WITH MASK and red boxes for WITHOUT MASK while triggering a beep and warning text.

Applications

Hospitals, airports, offices, schools, shopping malls, public transport, and industrial workplaces.
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Advantages and Limitations

Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency. Advantages include automation and real-time detection. Limitations include lighting sensitivity and dataset dependency.

Future Scope

Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices. Use YOLO-based face detection, larger datasets, mobile deployment, cloud integration, and edge devices.

Conclusion & References

The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection. The project successfully demonstrates CNN-based face mask detection. References include TensorFlow, OpenCV, Keras, and research articles on face mask detection.

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